EE2FH4
Electromagnetics I

Term II, January – April 2026

MATLAB Examples and Exercises (Set 3)

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Example: An infinite uniform linear charge $\rho_L = 2.0$ nC/m lies along the x axis in free space, while point charges of 8.0 nC each are located at (0, 0, 1) and (0, 0, -1). Find **E** at (2, 3, 4). Write a MATLAB program to verify your answer.

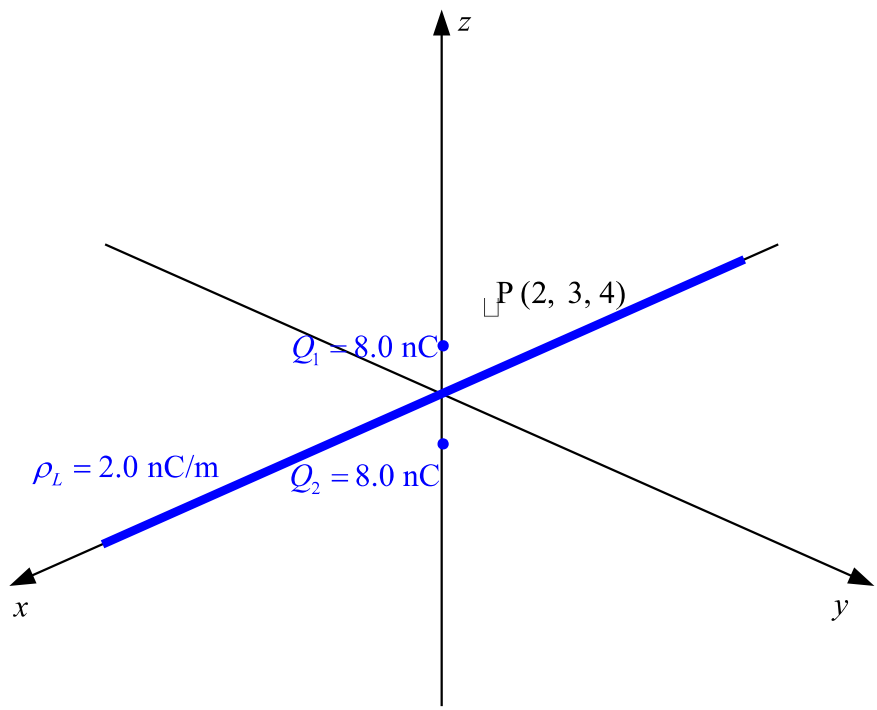

Figure 3.1 The charges of the example of Set 3.

Analytical Solution:

Based on the principle of superposition, the electric field at P (2, 3, 4) is $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_L$ where $\mathbf{E}_1$ and $\mathbf{E}_2$ are the electric fields generated by the point charges 1 and 2, respectively, and $\mathbf{E}_L$ is the electric field generated by the line charge. The electrical field generated by a point charge is given by

$$\mathbf{E}_{\text{point}} = \frac{Q}{4\pi\varepsilon_0 |\mathbf{R}|^3}\mathbf{R}$$

where $\mathbf{R}$ is the vector pointing from the point charge to the observation point as shown in Figure 3.2 (a).

For the point charge Q_1:

$$\mathbf{R_1} = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (\mathbf{a}_z) = 2\mathbf{a}_x + 3\mathbf{a}_y + 3\mathbf{a}_z$$

$$\mathbf{E}_1 = \frac{Q_1}{4\pi\varepsilon_0 |\mathbf{R_1}|^3}\mathbf{R_1}$$

$$= \frac{8\times10^{-9}}{4\pi\times\frac{1}{36\pi}\times10^{-9}\times\left(\sqrt{2^2+3^2+3^2}\right)^3}(2\mathbf{a}_x + 3\mathbf{a}_y + 3\mathbf{a}_z)$$

$$= 1.395\mathbf{a}_x + 2.093\mathbf{a}_y + 2.093\mathbf{a}_z$$

For the point charge Q_2:

$$\mathbf{R_2} = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (-\mathbf{a}_z) = 2\mathbf{a}_x + 3\mathbf{a}_y + 5\mathbf{a}_z$$

$$\mathbf{E}_2 = \frac{Q_2}{4\pi\varepsilon_0 |\mathbf{R_2}|^3}\mathbf{R_2}$$

$$= \frac{8\times10^{-9}}{4\pi\times\frac{1}{36\pi}\times10^{-9}\times\left(\sqrt{2^2+3^2+5^2}\right)^3}(2\mathbf{a}_x + 3\mathbf{a}_y + 5\mathbf{a}_z)$$

$$= 0.615\mathbf{a}_x + 0.922\mathbf{a}_y + 1.537\mathbf{a}_z$$

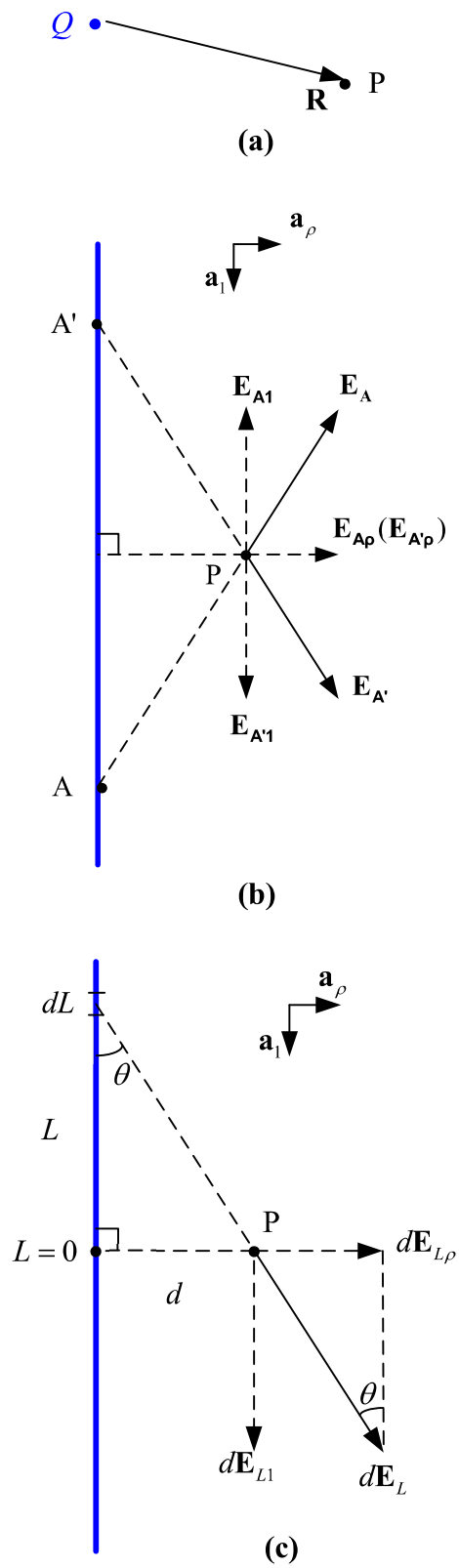

Figure 3.2. The different field components

As we can see from Figure 3.2(b), for any point A on the line charge we can always find one and only one point A' whose electric field at P has the same magnitude but the opposite sign of that of A in the direction which is parallel to the line charge. This is because the linear charge is infinitely long. Therefore we only need to find the electric field in the direction perpendicular to the line charge. As shown in Figure 3.2(c) each incremental length of line charge dL acts as a point charge and produces an incremental contribution to the total electric field intensity. The magnitude of $d\mathbf{E_L}$ is thus:

$$dE_L = \frac{dQ}{4\pi\varepsilon_0 |\mathbf{R}|^2} = \frac{\rho_L dL}{4\pi\varepsilon_0 (L^2 + d^2)}$$

therefore the magnitude of $d\mathbf{E}_\rho$ is

$$dE_{L\rho} = dE_L \sin\theta = \frac{\rho_L dL}{4\pi\varepsilon_0 (L^2 + d^2)} \frac{d}{\sqrt{(L^2 + d^2)}} = \frac{\rho_L d}{4\pi\varepsilon_0} \frac{dL}{(L^2 + d^2)^{3/2}}$$

and $E_{L\rho} = \int_{L=-\infty}^{L=\infty} dE_\rho \quad = \frac{\rho_L d}{4\pi\varepsilon_0} \int_{L=-\infty}^{L=\infty} \frac{dL}{(L^2 + d^2)^{3/2}}$ (this integral is given in the formula sheet)

$$E_{L\rho} = \frac{\rho_L d}{4\pi\varepsilon_0} \times \left. \frac{L}{d^2\sqrt{L^2 + d^2}} \right|_{L=-\infty}^{L=\infty}$$

$$= \frac{\rho_L d}{4\pi\varepsilon_0} \times \left(\left. \frac{1}{d^2\sqrt{1 + \frac{d^2}{L^2}}} \right|_{L=\infty} - \left. \frac{-1}{d^2\sqrt{1 + \frac{d^2}{L^2}}} \right|_{L=-\infty} \right) = \frac{\rho_L d}{4\pi\varepsilon_0} \times \left(\frac{1}{d^2\sqrt{1+0}} - \frac{-1}{d^2\sqrt{1+0}} \right) = \frac{\rho_L}{2\pi\varepsilon_0 d}.$$

Then we have to find $\mathbf{a}_\rho$

$$\because \mathbf{R}_\rho = (2\mathbf{a}_x + 3\mathbf{a}_y + 4\mathbf{a}_z) - (2\mathbf{a}_x) = 3\mathbf{a}_y + 4\mathbf{a}_z$$

$$\therefore \mathbf{a}_\rho = \frac{\mathbf{R}_\rho}{|\mathbf{R}_\rho|} = \frac{3\mathbf{a}_y + 4\mathbf{a}_z}{\sqrt{3^2 + 4^2}} = \frac{3}{5}\mathbf{a}_y + \frac{4}{5}\mathbf{a}_z$$

therefore

$$\mathbf{E_L} = \mathbf{E}_{L\rho} = E_{L\rho}\mathbf{a}_\rho$$

$$= \frac{2\times10^{-9}}{2\pi \times \frac{1}{36\pi} \times 10^{-9} \times 5} \times \left(\frac{3}{5}\mathbf{a}_y + \frac{4}{5}\mathbf{a}_z \right)$$

$$= 4.32\mathbf{a}_y + 5.76\mathbf{a}_z$$

and

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E_L} = 2.01\mathbf{a}_x + 7.34\mathbf{a}_y + 9.39\mathbf{a}_z \text{ V/m}$$

MATLAB solution :

To find the electric field generated by the infinite line charge, we can replace the infinite linear charge by a sufficiently long finite line charge. In this problem, our line charge has a length of one hundred times of the distance from the observation point to the line charge, and its center is located at C (2, 0, 0) as shown in Figure 3.3. We divide the line charge into many equal segments, and evaluate the electric field generated by each segment in the way we evaluate the electric field generated by a single point charge. Finally, we evaluate the summation of the electric fields generated by all those segments. The summation should be very close to the electric field generated by the infinite linear charge. This approach can be summarized by the mathematical expression:

$$\mathbf{E_L} \approx \sum_{i=1}^{n} \mathbf{E}_i = \sum_{i=1}^{n} \frac{\Delta Q}{4\pi\varepsilon_0 |\mathbf{R}_i|^3}\mathbf{R}_i = \sum_{i=1}^{n} \frac{\rho_L \Delta L}{4\pi\varepsilon_0 |\mathbf{R}_i|^3}\mathbf{R}_i$$

where $\mathbf{E}_i$ is the electric field generated by the *i*th segment, $\mathbf{R}_i$ is the vector from the *i*th segment to the observation point, ΔQ is the charge of a single segment, ΔL is the length of the segment and *n* is the total number of the segments.

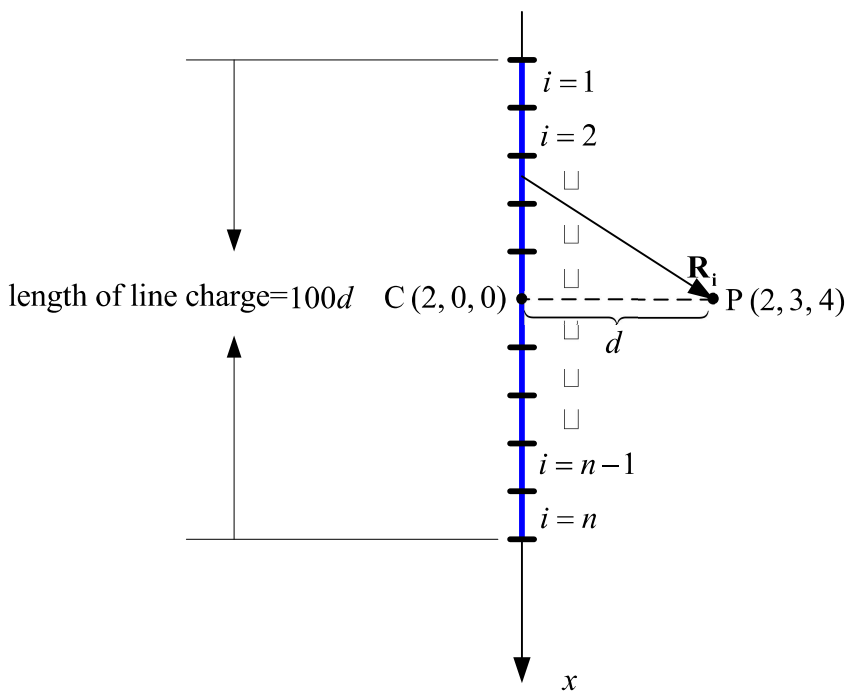

Figure 3.3 The discretization used in the MATLAB program.

MATLAB code :

```
clc; %clear the command line
clear; %remove all previous variables
Q1=8e-9;%charges on Q1
Q2=8e-9;%charges on Q2
pL=2e-9;%charge density of the line
Epsilono=8.8419e-12;%Permitivity of free space

P=[2 3 4];%coordinates of observation point
A=[0 0 1];%coordinates of Q1
B=[0 0 -1];%coordinates of Q2
C=[2 0 0];%coordinates of the center of the line charge
Number_of_L_Steps=100000;%the steps of L

%%the following routine calculates the electric fields at the
%%observation point generated by the point charges
R1=P-A; %the vector pointing from Q1 to the observation point
R2=P-B; %the vector pointing from Q2 to the observation point
R1Mag=norm(R1);%the magnitude of R1
R2Mag=norm(R2);%the magnitude of R1
E1=Q1/(4*pi*Epsilono*R1Mag^3)*R1;%the electric field generated by Q1
E2=Q2/(4*pi*Epsilono*R2Mag^3)*R2;%the electric field generated by Q2

%%the following routine calculates the electric field at the
%%observation point generated by the line charge
d=norm(P-C);%the distance from the observation point to the center of the line
length=100*d;%the length of the line
dL_V=length/Number_of_L_Steps*[1 0 0];%vector of a segment
dL=norm(dL_V);%length of a segment
EL=[0 0 0];%initialize the electric field generated by EL
C_segment=C-( Number_of_L_Steps/2*dL_V-dL_V/2);%the center of the first segment
for i=1: Number_of_L_Steps
    R=P-C_segment;%the vector seen from the center of the first segment to the
observation point
    RMag=norm(R);%the magnitude of the vector R
    EL=EL+dL*pL/(4*pi*Epsilono*RMag^3)*R;%get contibution from each segment
    C_segment=C_segment+dL_V;%the center of the i-th segment
end

E=E1+E2+EL;% the electric field at P
```

Running result

```
Command Window
>> E

E =

    2.0102    7.3345    9.3890

>>
```

Comparing the MATLAB answer with the analytical answer we see that there is very little difference between them. This little difference is caused by the finite length of the steps of L and the finite line charge that we used to replace the infinite one.

Exercise: A finite uniform linear charge $\rho_L = 4$ nC/m lies on the xy plane as shown in Figure 3.4, while point charges of 8 nC each are located at (0, 1, 1) and (0, -1, 1). Find **E** at (0, 0 ,0). Write a MATLAB program to verify your answer.

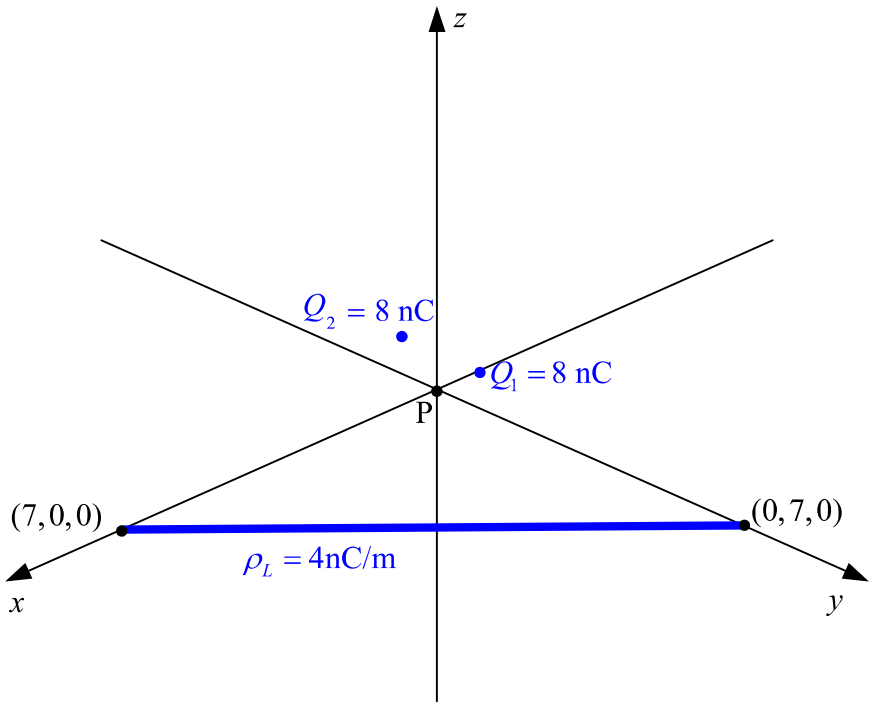

Figure 3.4 The charges of the exercise of Set 3.